

# JEE MAIN REVISION MASTER: CAPACITOR 2026 2 QUESTIONS 3.33% WEIGHTAGE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 17 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                      | Frequency | Category        |
|---------------|-----------------------------------|-----------|-----------------|
| Very High     | Kcet Capacitor - Core Application | 17 PYQs   | General Concept |

## 2. Master Formula Reference Card

| Concept / Topic | Mathematical Formulation & Governing Equations |
|-----------------|------------------------------------------------|
|-----------------|------------------------------------------------|

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Kcet Capacitor - Core Application (Very High Yield — 17 PYQs)

Core Principle: Key problem pattern in Kcet Capacitor.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the circuit shown in the figure, the potential difference across the  $4\mu F$  capacitor is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "A parallel plate capacitor has a uniform electric field 'E' in the space between the plates. If the distance between the plates is 'd...' [Ans: C]
- [2024 Shift] Q#3 (MCQ): "A capacitor of capacitance  $5\mu F$  is charged by a battery of emf 10 V. At an instant of time, the potential difference across the capacit..." [Ans: N/A]

## 4. Benchmark Illustrative PYQs with Master Solutions

**Benchmark Problem 1: KCET 2026 (Q#1)**

In the circuit shown in the figure, the potential difference across the  $4\mu\text{F}$  capacitor is

- (A) 3 V
- (B) 4 V
- (C) 9 V
- (D) 12 V

**Correct Answer: None**

**Solution Analysis:** The  $9\mu\text{F}$  and  $3\mu\text{F}$  capacitors are connected in parallel. In parallel combination, capacitances are added directly.  $C_{(p)} = 9 + 3 = 12\mu\text{F}$  Now this  $12\mu\text{F}$  equivalent capacitor is in series with the  $4\mu\text{F}$  capacitor. For two capacitors in series, the equivalent capacitance is:  $C_{(eq)} = \frac{4 \times 12}{4 + 12} = \frac{48}{16} = 3\mu\text{F}$  The battery voltage is 12V. So the total charge supplied by the battery is:  $Q = C_{(eq)} \times V = 3\mu\text{F} \times 12\text{V} = 36\mu\text{C}$  In a series combination, each capacitor has the same charge. So the charge